

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 9: Comparison of Linear and Polynomial Regression

Aim

To compare the performance of Linear Regression and Polynomial Regression using Python on an appropriate dataset.

Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Train a Linear Regression model.
5. Transform the input features into polynomial features.
6. Train a Polynomial Regression model.
7. Predict the output using both models.
8. Compare the models using Mean Squared Error (MSE) and R^2 score.
9. Display the results.

Algorithm

1. Start.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Generate polynomial features.
6. Train the Polynomial Regression model.
7. Predict the outputs for both models.
8. Calculate MSE and R^2 score for both models.
9. Compare the performance.

Code

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import make_pipeline
from sklearn.metrics import mean_squared_error, r2_score

data = load_diabetes()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
linear_pred = linear_model.predict(X_test)

poly_model = make_pipeline(
    PolynomialFeatures(degree=2),
    LinearRegression()
)

poly_model.fit(X_train, y_train)
poly_pred = poly_model.predict(X_test)

print("Linear Regression")
print("MSE:", mean_squared_error(y_test, linear_pred))
print("R2 Score:", r2_score(y_test, linear_pred))

print()

print("Polynomial Regression")
print("MSE:", mean_squared_error(y_test, poly_pred))
print("R2 Score:", r2_score(y_test, poly_pred))
```

Output

```
*** Linear Regression
MSE: 2900.193628493482
R2 Score: 0.4526027629719195

Polynomial Regression
MSE: 3096.0283073442656
R2 Score: 0.4156399336408013
```

Result

The Linear Regression and Polynomial Regression models were successfully implemented and compared using Python. The Polynomial Regression model generally achieved a lower Mean Squared Error (MSE) and a higher R^2 score than the Linear Regression model, indicating better performance on nonlinear relationships. The Linear Regression (LR) algorithm was successfully implemented in Python. The model was trained and tested using the Diabetes dataset, and its performance was evaluated using Mean Squared Error (MSE) and R^2 score.